COMPUTER ENGINEERING

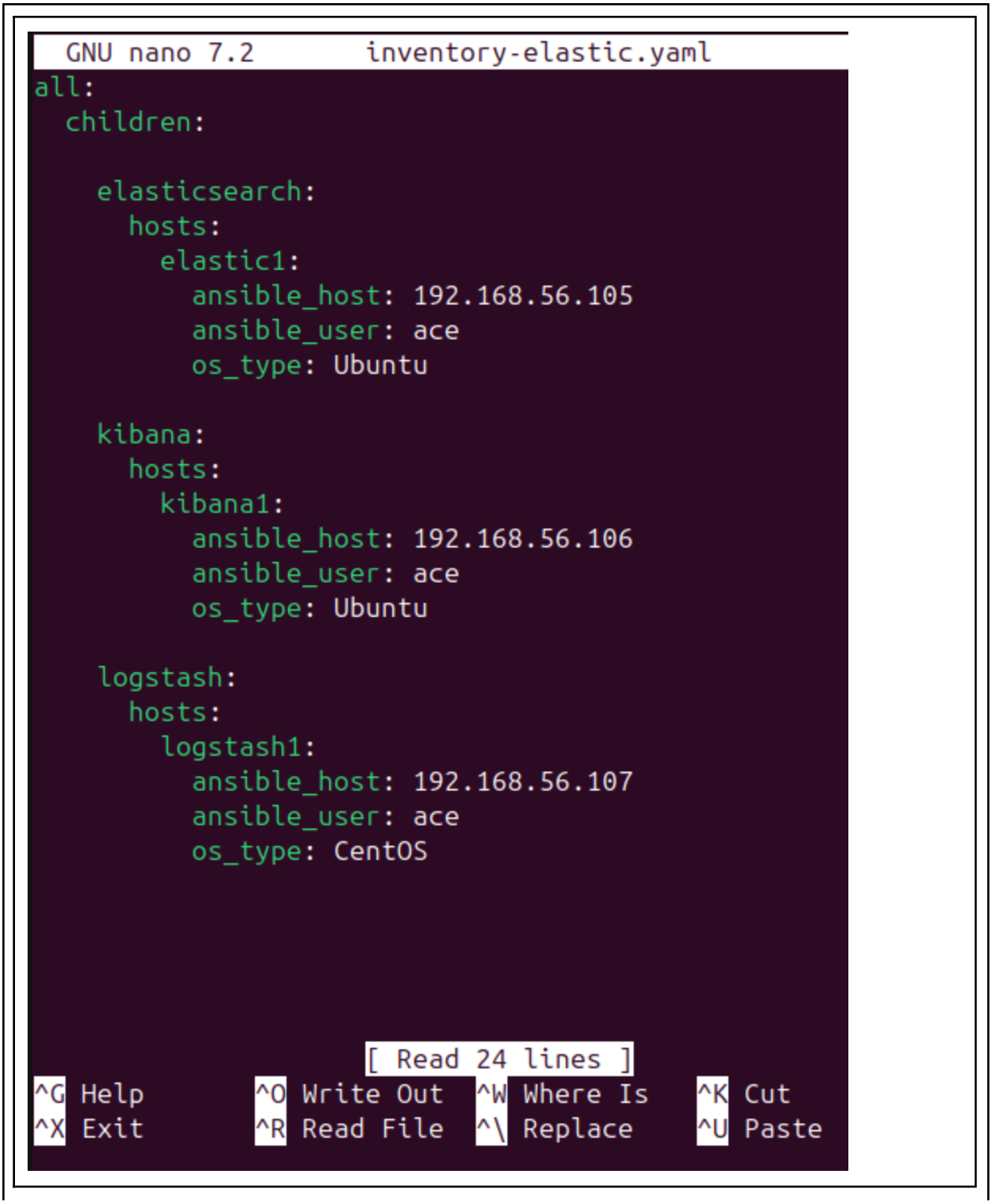
QUEZON CITY

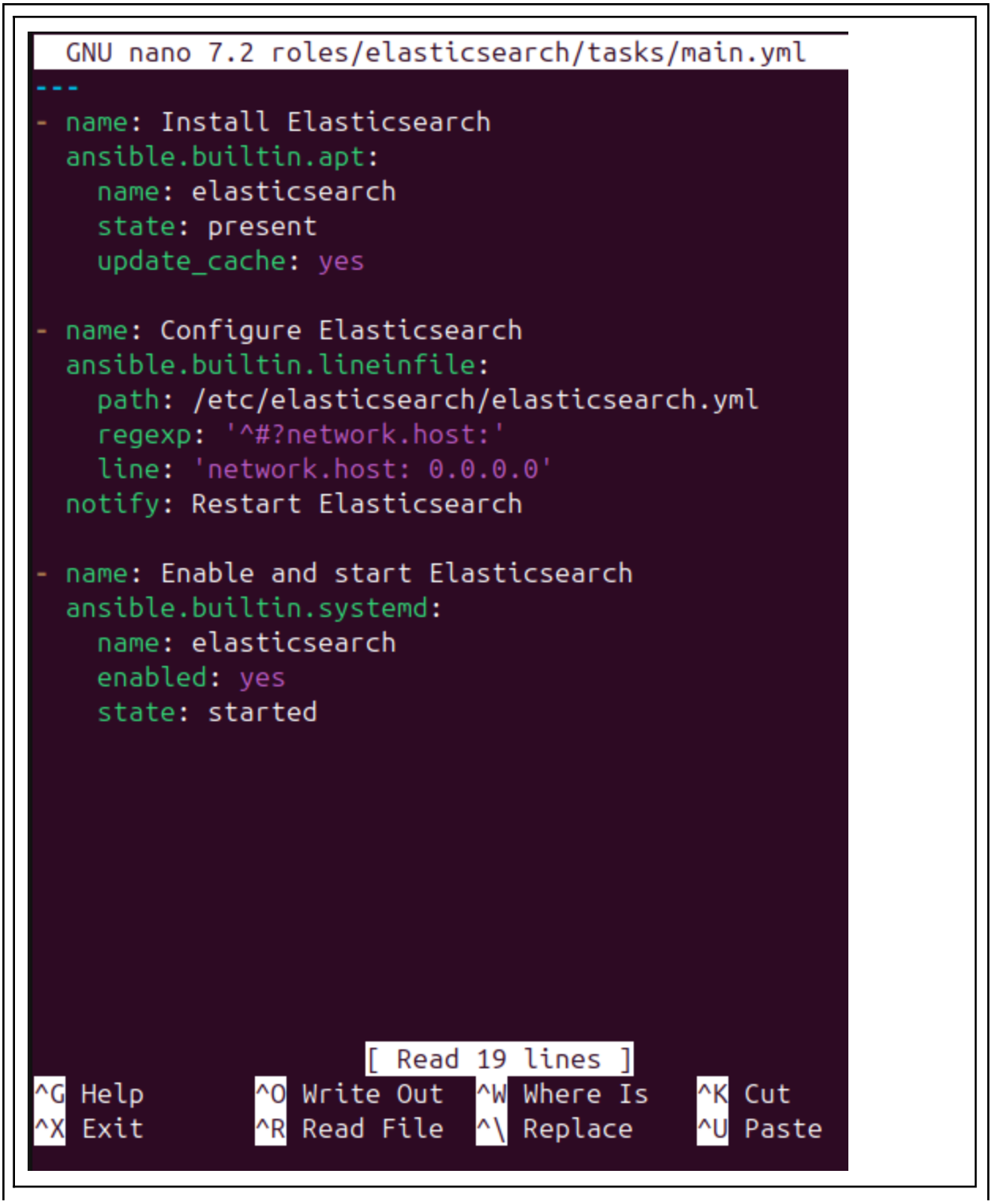
COURSE CODE/TITLE:	CPE212	**SECTION:**	CPE31S2
DATE PERFORMED:		**DATE SUBMITTED:**	
NAME:	Ace Tadalan	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. 10 **Elastic Stack Ansible Configuration**

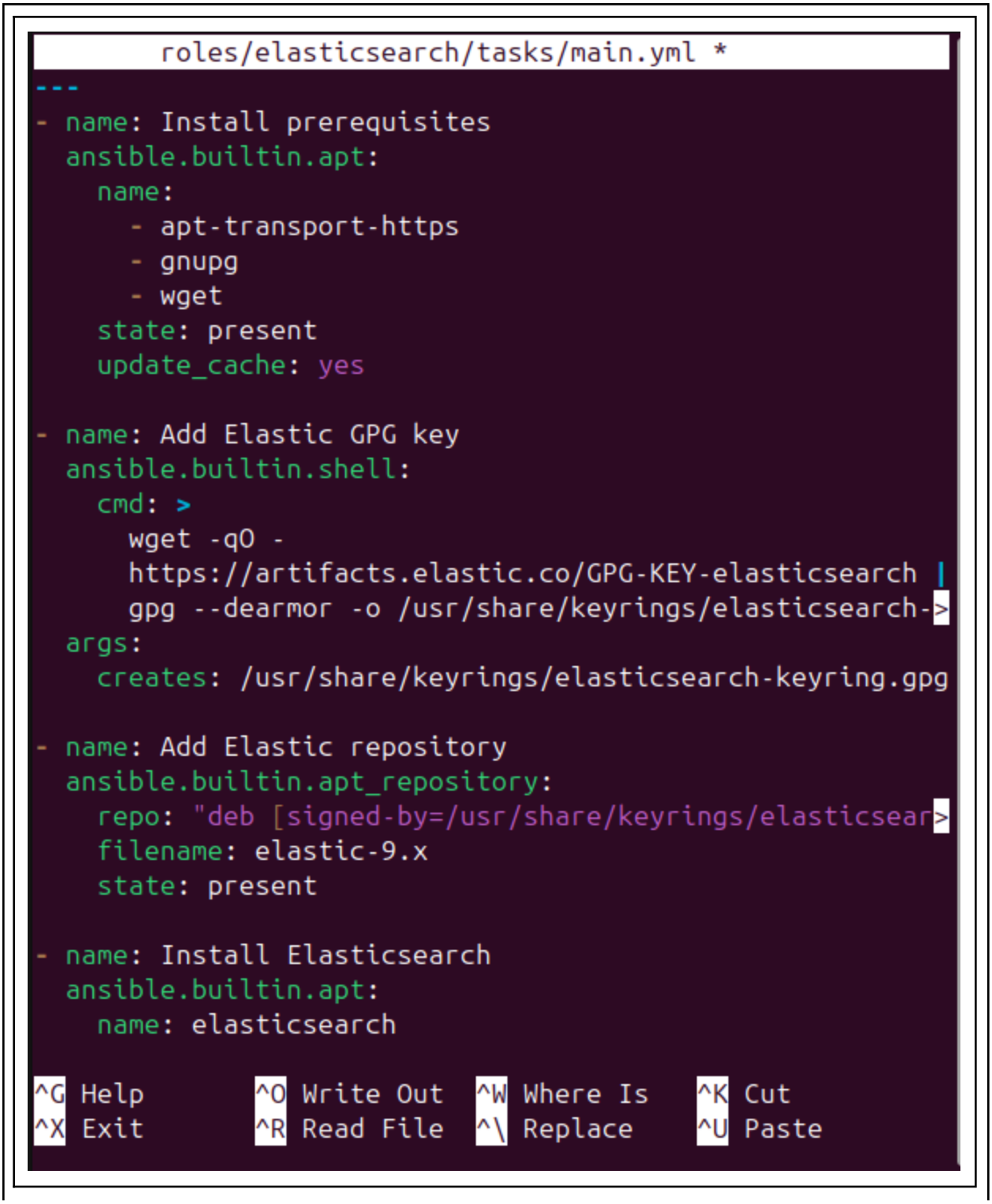
1. PROCEDURE OUTPUT

1. Create a playbook that:
    a. Install and configure Elastic Stack in separate hosts (Elastic Search, Kibana, Logstash)
2. Apply the concept of creating roles.
3. Describe how you did step 1. (Provide screenshots and explanations in your report. Make your report detailed such that it will look like a manual.)
4. Show an output of the installed Elastic Stack for both Ubuntu and CentOS.
5. Make sure to create a new repository in GitHub for this activity.

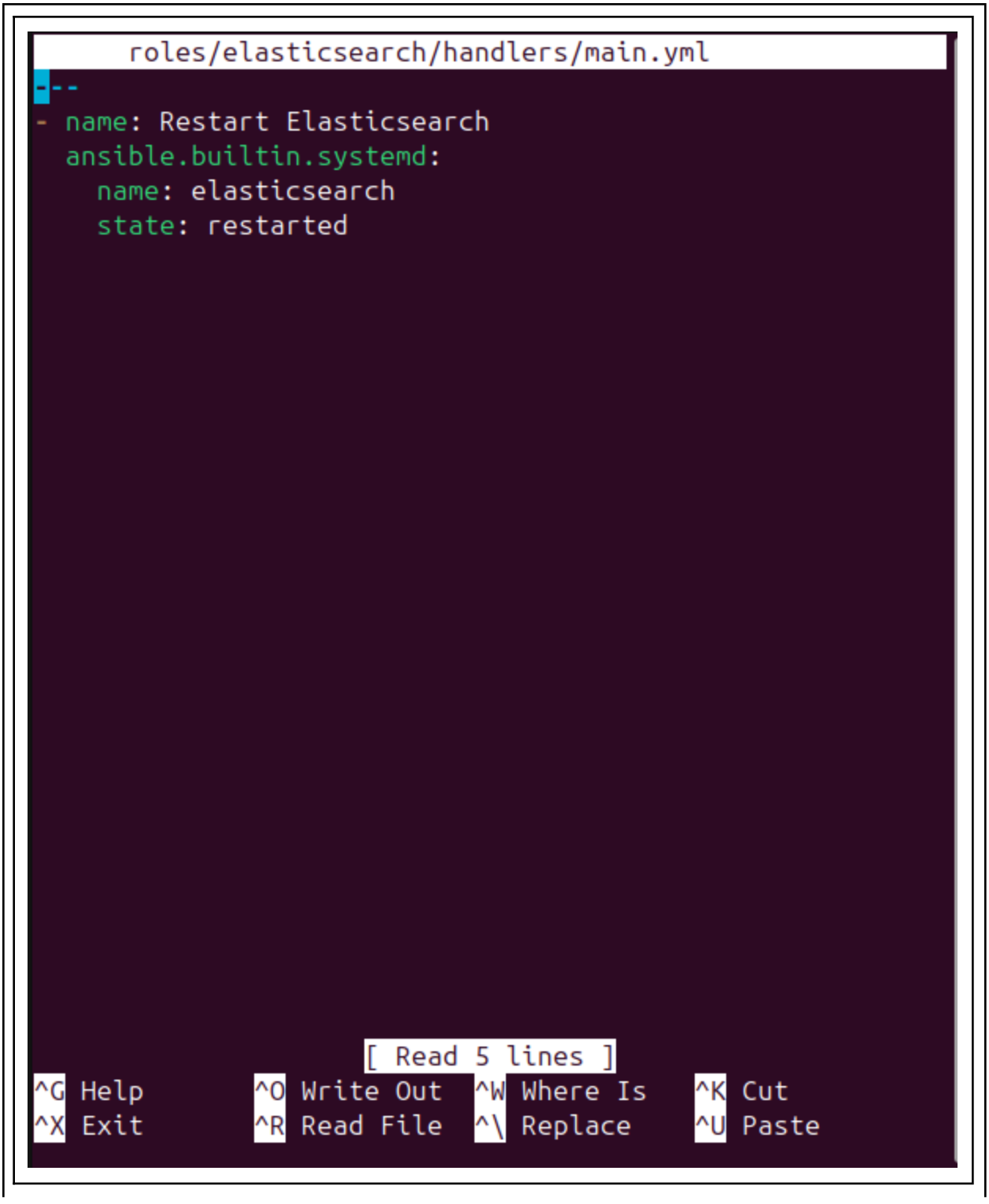

GNU nano 7.2 inventory-elastic.yaml
all:
children:
elasticsearch:
hosts:
elastic1:
ansible_host: 192.168.56.105
ansible_user: ace
os_type: Ubuntu
kibana:
hosts:
kibana1:
ansible_host: 192.168.56.106
ansible_user: ace
os_type: Ubuntu
logstash:
hosts:
logstash1:
ansible_host: 192.168.56.107
ansible_user: ace
os_type: CentOS
[Read 24 lines]
^G Help ^O Write Out ^W Where Is ^K Cut
^X Exit ^R Read File ^\ Replace ^U Paste


GNU nano 7.2 roles/elasticsearch/tasks/main.yml

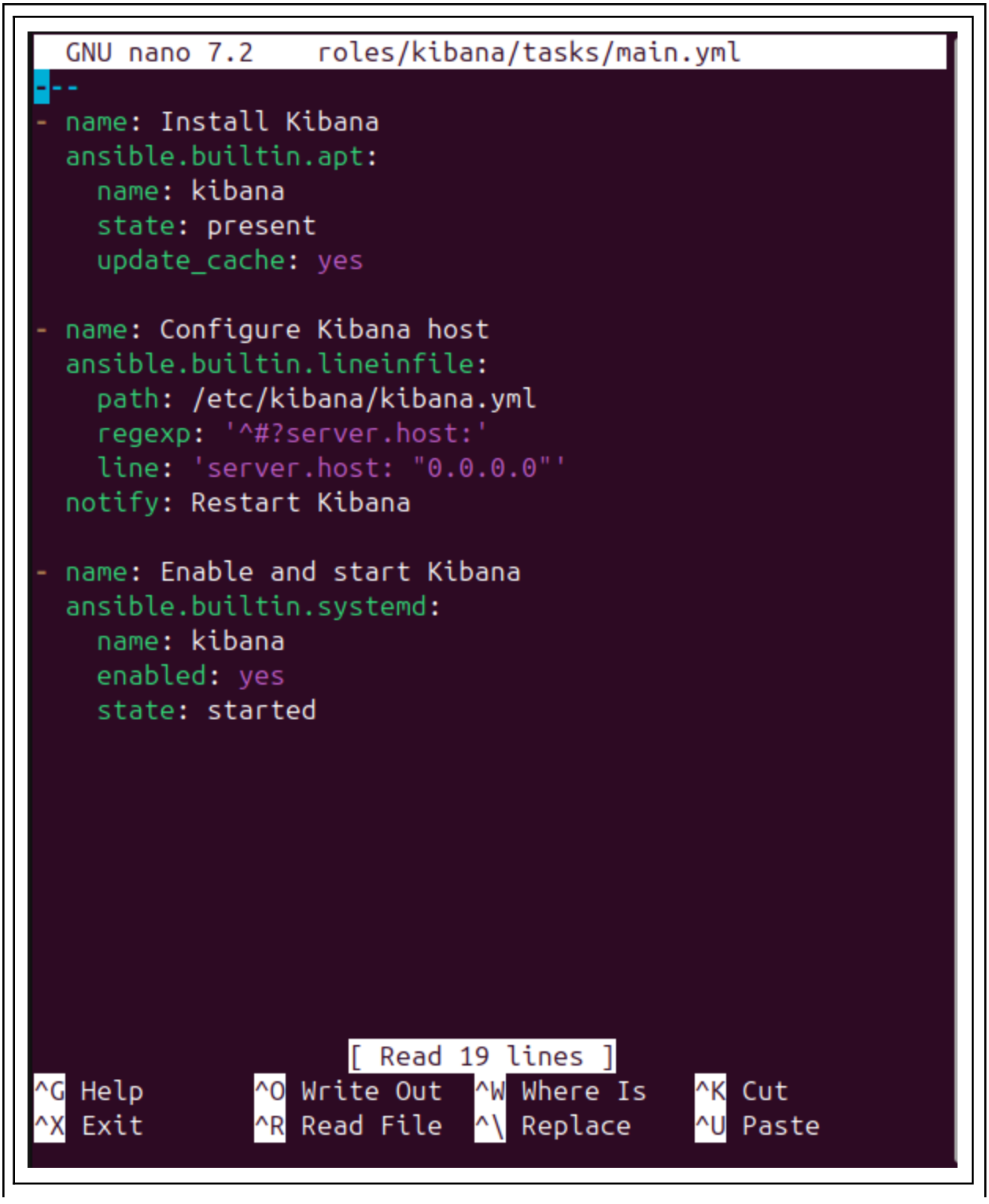
- name: Install Elasticsearch
ansible.builtin.apt:
name: elasticsearch
state: present
update_cache: yes
- name: Configure Elasticsearch
ansible.builtin.lineinfile:
path: /etc/elasticsearch/elasticsearch.yml
regexp: '^#?network.host:'
line: 'network.host: 0.0.0.0'
notify: Restart Elasticsearch
- name: Enable and start Elasticsearch
ansible.builtin.systemd:
name: elasticsearch
enabled: yes
state: started
[Read 19 lines]
^G Help
^O Write Out
^W Where Is
^K Cut
^X Exit
^R Read File
^\ Replace
^U Paste


roles/elasticsearch/tasks/main.yml *

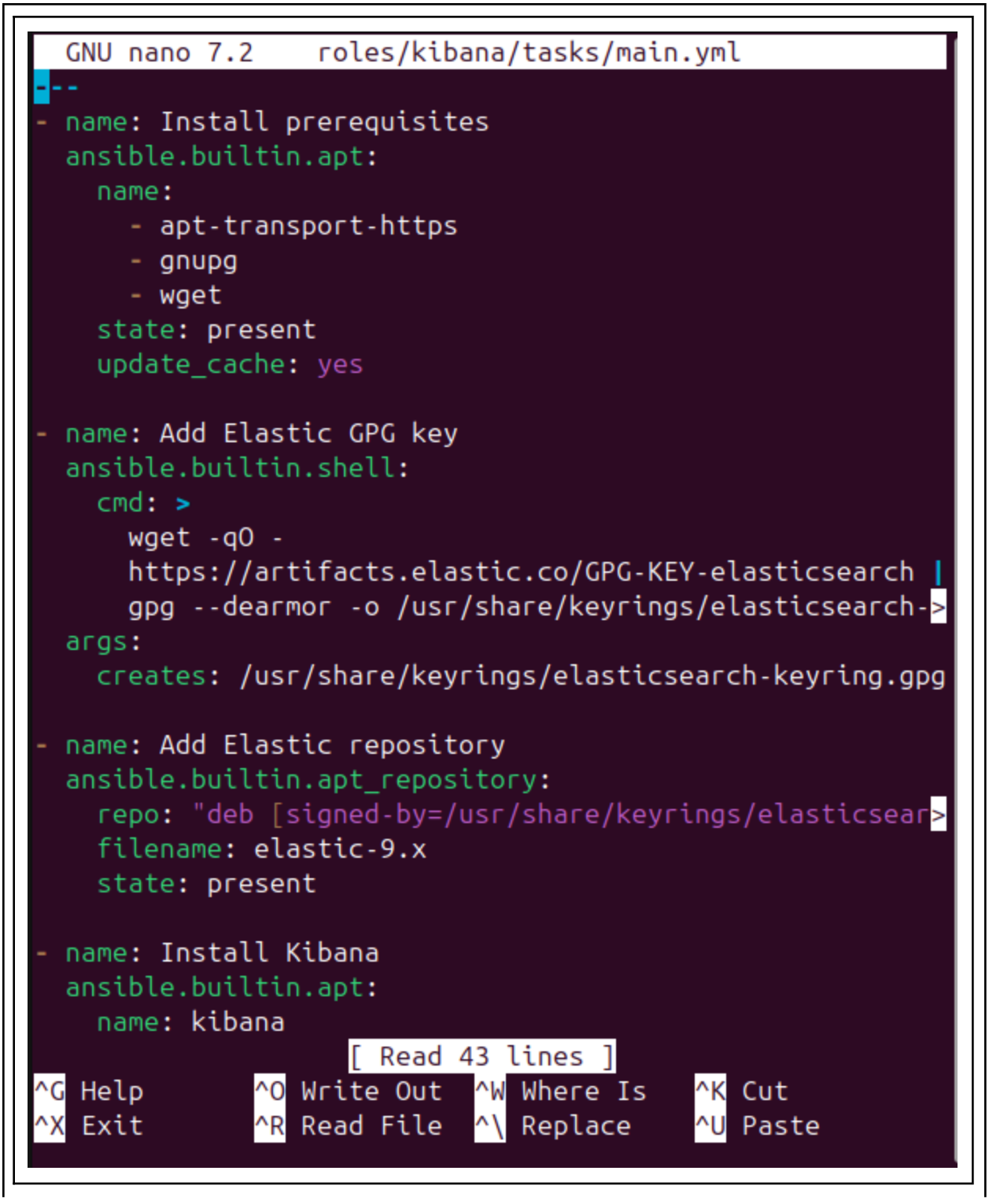
- name: Install prerequisites
ansible.builtin.apt:
name:
- apt-transport-https
- gnupg
- wget
state: present
update_cache: yes
- name: Add Elastic GPG key
ansible.builtin.shell:
cmd: >
wget -qO -
https://artifacts.elastic.co/GPG-KEY-elasticsearch |
gpg --dearmor -o /usr/share/keyrings/elasticsearch-
args:
creates: /usr/share/keyrings/elasticsearch-keyring.gpg
- name: Add Elastic repository
ansible.builtin.apt_repository:
repo: "deb [signed-by=/usr/share/keyrings/elasticsear
filename: elastic-9.x
state: present
- name: Install Elasticsearch
ansible.builtin.apt:
name: elasticsearch
^G Help ^O Write Out ^W Where Is ^K Cut
^X Exit ^R Read File ^\ Replace ^U Paste


roles/elasticsearch/handlers/main.yml

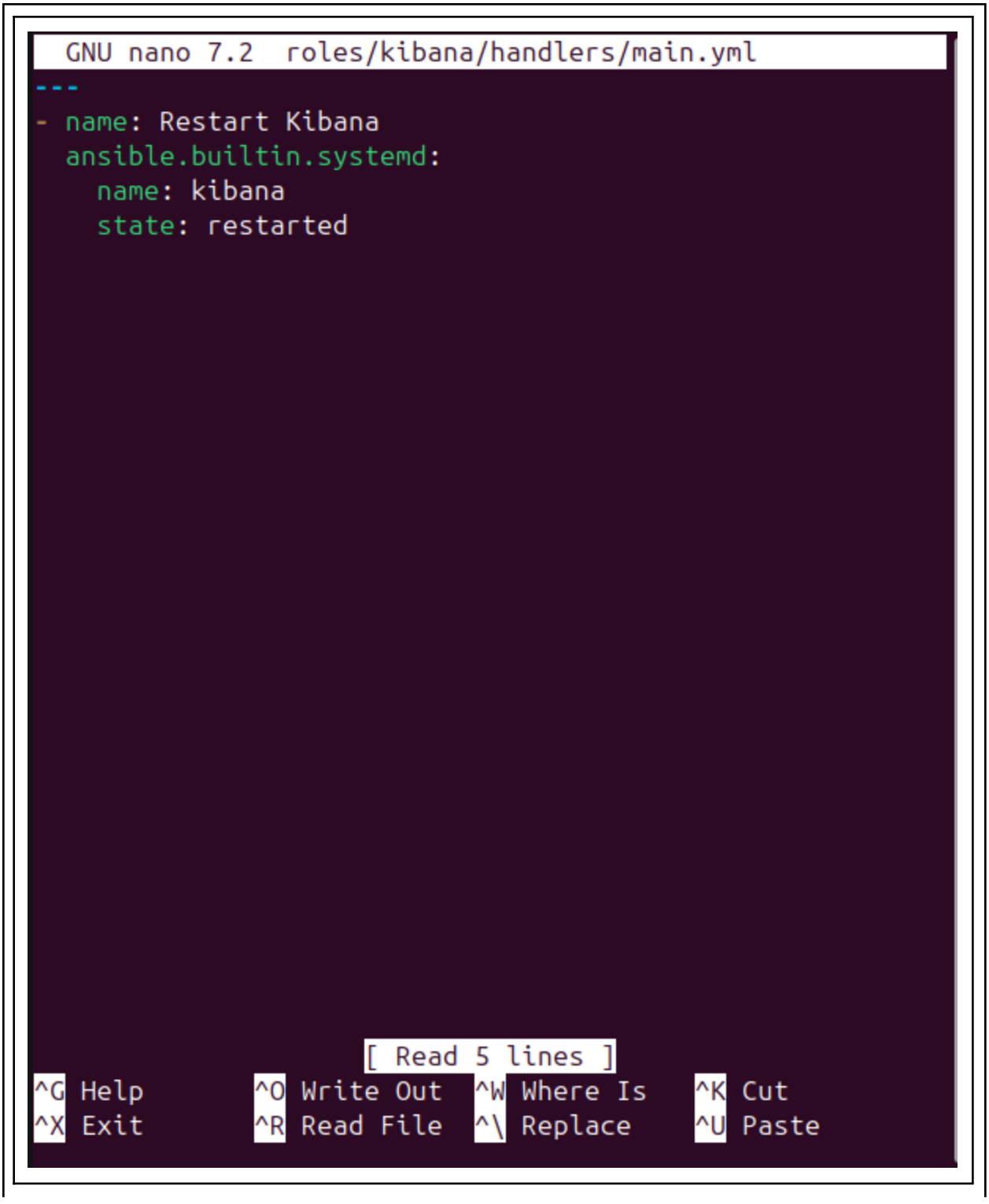
- name: Restart Elasticsearch
ansible.builtin.systemd:
name: elasticsearch
state: restarted
[Read 5 lines]
^G Help
^O Write Out
^W Where Is
^K Cut
^X Exit
^R Read File
^\ Replace
^U Paste


GNU nano 7.2 roles/kibana/tasks/main.yml

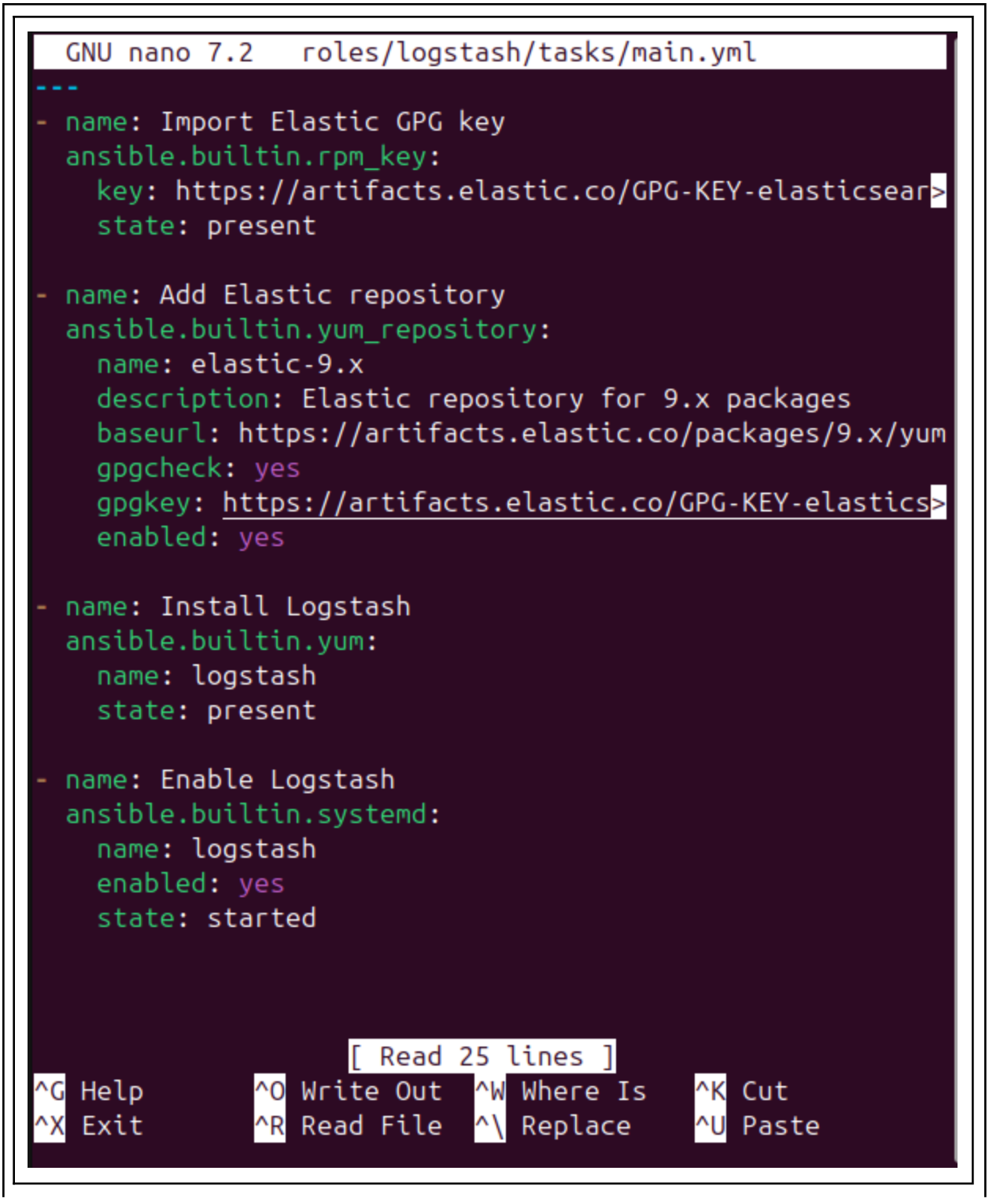
- name: Install Kibana
ansible.builtin.apt:
name: kibana
state: present
update_cache: yes
- name: Configure Kibana host
ansible.builtin.lineinfile:
path: /etc/kibana/kibana.yml
regexp: '^#?server.host:'
line: 'server.host: "0.0.0.0"'
notify: Restart Kibana
- name: Enable and start Kibana
ansible.builtin.systemd:
name: kibana
enabled: yes
state: started
[Read 19 lines]
^G Help ^O Write Out ^W Where Is ^K Cut
^X Exit ^R Read File ^\ Replace ^U Paste


GNU nano 7.2 roles/kibana/tasks/main.yml

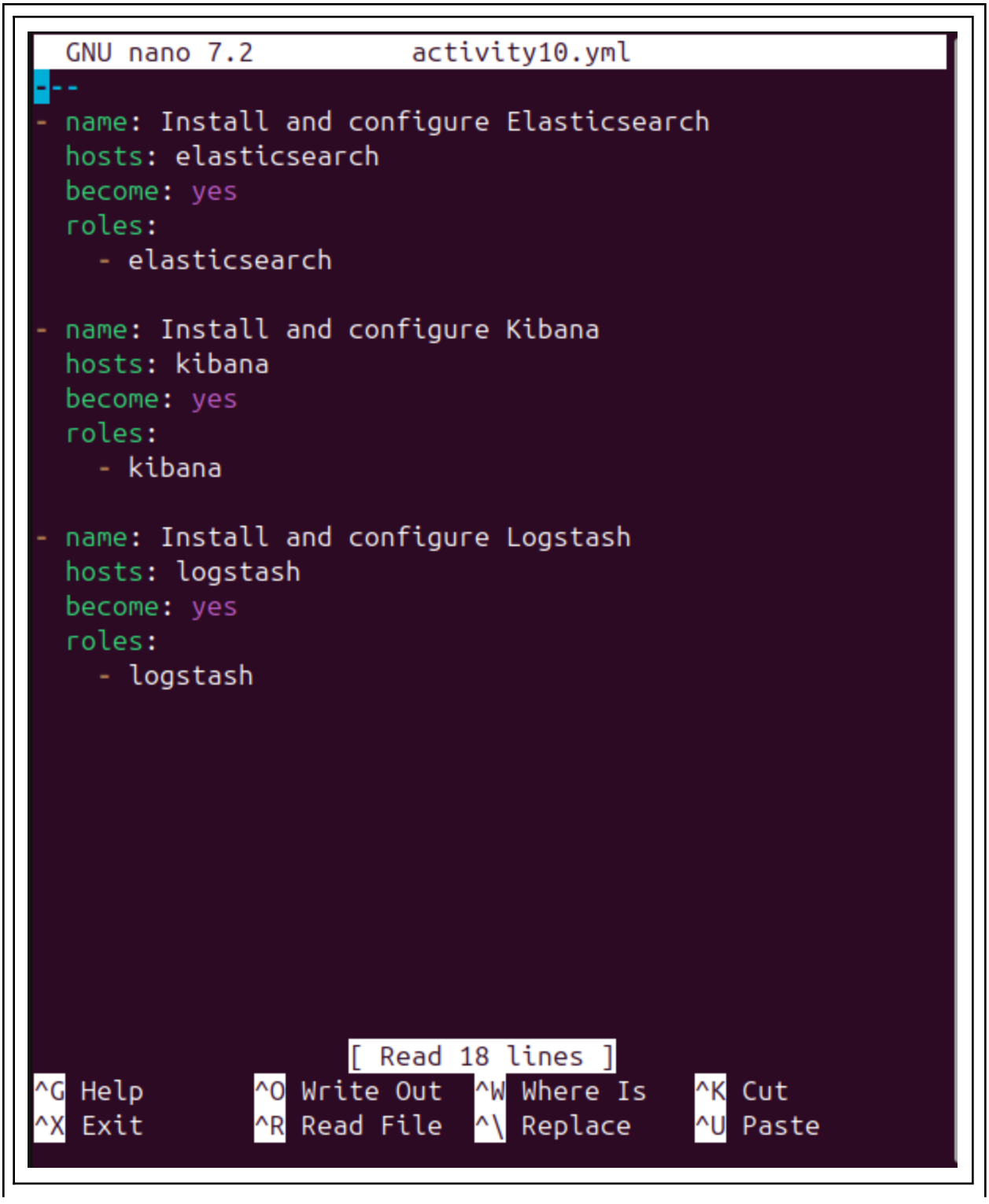
- name: Install prerequisites
ansible.builtin.apt:
name:
- apt-transport-https
- gnupg
- wget
state: present
update_cache: yes
- name: Add Elastic GPG key
ansible.builtin.shell:
cmd: >
wget -qO -
https://artifacts.elastic.co/GPG-KEY-elasticsearch |
gpg --dearmor -o /usr/share/keyrings/elasticsearch-
args:
creates: /usr/share/keyrings/elasticsearch-keyring.gpg
- name: Add Elastic repository
ansible.builtin.apt_repository:
repo: "deb [signed-by=/usr/share/keyrings/elasticsear
filename: elastic-9.x
state: present
- name: Install Kibana
ansible.builtin.apt:
name: kibana
[Read 43 lines]
^G Help ^O Write Out ^W Where Is ^K Cut
^X Exit ^R Read File ^\ Replace ^U Paste


GNU nano 7.2 roles/kibana/handlers/main.yml

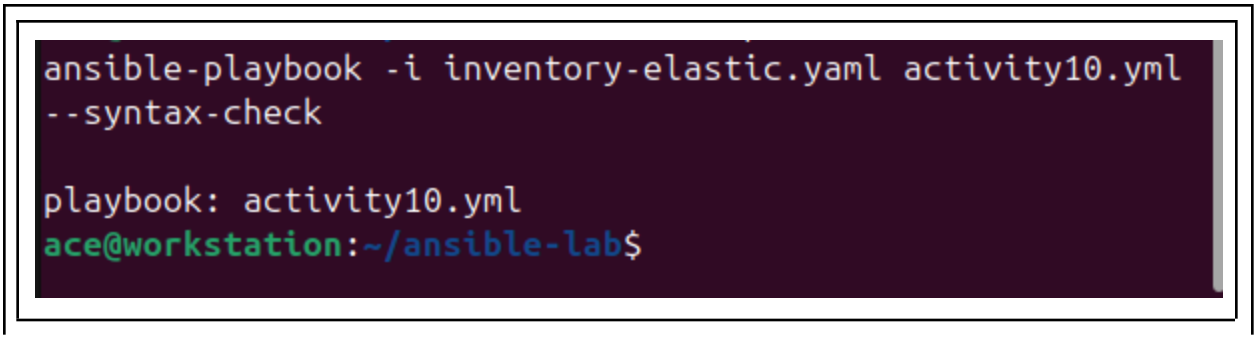
- name: Restart Kibana
ansible.builtin.systemd:
name: kibana
state: restarted
[Read 5 lines]
^G Help
^O Write Out
^W Where Is
^K Cut
^X Exit
^R Read File
^\ Replace
^U Paste

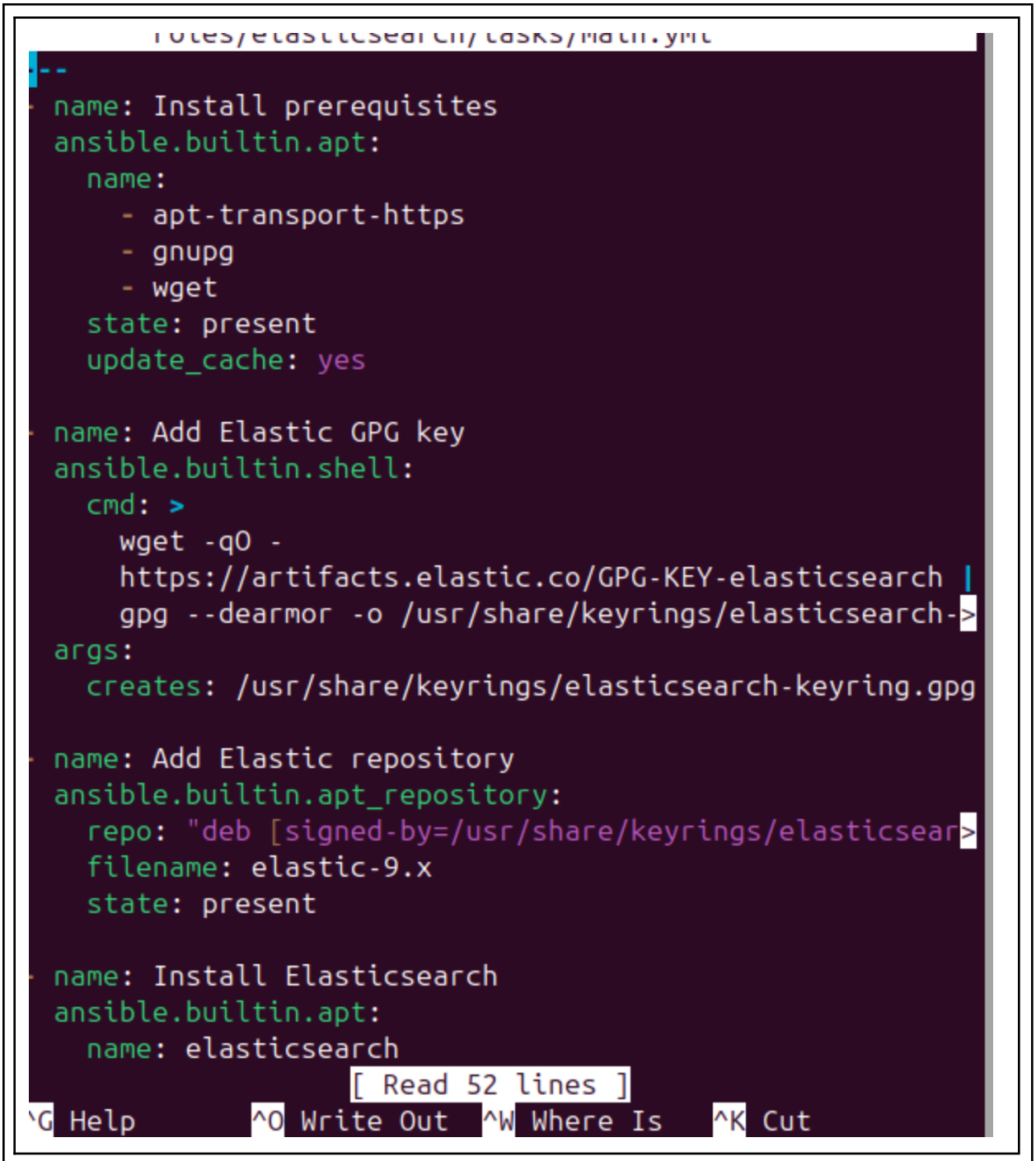

GNU nano 7.2 roles/logstash/tasks/main.yml

- name: Import Elastic GPG key
ansible.builtin.rpm_key:
key: https://artifacts.elastic.co/GPG-KEY-elasticsear>
state: present
- name: Add Elastic repository
ansible.builtin.yum_repository:
name: elastic-9.x
description: Elastic repository for 9.x packages
baseurl: https://artifacts.elastic.co/packages/9.x/yum
gpgcheck: yes
gpgkey: https://artifacts.elastic.co/GPG-KEY-elastics>
enabled: yes
- name: Install Logstash
ansible.builtin.yum:
name: logstash
state: present
- name: Enable Logstash
ansible.builtin.systemd:
name: logstash
enabled: yes
state: started
[Read 25 lines]
^G Help ^O Write Out ^W Where Is ^K Cut
^X Exit ^R Read File ^\ Replace ^U Paste


GNU nano 7.2 activity10.yml

- name: Install and configure Elasticsearch
hosts: elasticsearch
become: yes
roles:
- elasticsearch
- name: Install and configure Kibana
hosts: kibana
become: yes
roles:
- kibana
- name: Install and configure Logstash
hosts: logstash
become: yes
roles:
- logstash
[Read 18 lines]
^G Help ^O Write Out ^W Where Is ^K Cut
^X Exit ^R Read File ^\ Replace ^U Paste


ansible-playbook -i inventory-elastic.yaml activity10.yml
--syntax-check
playbook: activity10.yml
ace@workstation:~/ansible-lab$



Reflections:

Answer the following:

1. What are the benefits of having log monitoring tool?
   - Log monitoring tools mainly used to alert users when there are noticeable circumstances occur, that directly helping administrators respond to potential

problems in terms of security issues as well as it benefits an organization, as they continuously monitor server, application, and network logs to early identify irregularities and unusual patterns.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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